

Georgia Channing

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EDUCATION

University of Oxford

DPhil. Computer Science, AIMS CDT

Oxford, UK

Oct. 2024 – Apr. 2027

University of Oxford

MSc. Advanced Computer Science, Google DeepMind Scholar

Oxford, UK

Oct. 2023 – Sep. 2024

University of Tennessee

BSc. Computer Science, *summa cum laude*

Knoxville, TN

Aug. 2017 – May 2022

ETH Zürich

Informatik Bachelor's Thesis, 5.75/6.0

Zürich, CH

Feb. 2019 – Jul. 2019

EXPERIENCE

Global Computing Lab

Staff Researcher

Jun. 2022 – Oct. 2023

Knoxville, TN

- Primary researcher on NSF-funded project developing tools to analyze neural networks and optimize neural architecture search workflows with Pytorch, Ray, Weights & Biases, and Google Cloud Services.
- Analyzed and visualized performance of different neural architectures across benchmark datasets. Developed neural architectures that minimize energy consumption.
- Collaborated with chemists and physicists at RIKEN. Recruited and managed research students.

Publications: [Google Scholar](#).

Center for Advanced Defense Studies (C4ADS)

Data Scientist

Oct. 2020 – May 2022

Washington, D.C.

- Built computer vision tools with satellite imagery and AIS signals to identify illegal vessel behavior, proliferation of nuclear precursors, and human trafficking.
- Managed the data collection pipeline and its support team of nine data science consultants and two interns.
- Maintained and developed a proprietary datalake with terabytes of data and more than 150 individual datasets with AWS.
- Wrote dossiers and reports for distribution to partners, governments, and the public.

Highlights: [Sanctions Explorer](#), [Signals in the Noise](#), [Open Gates](#).

PROJECTS

Master's Thesis | *Towards Robust, Explainable Deepfake Audio Detection*

Apr. 2024 – Sep. 2024

- Developed multi-modal tools for audio explainability, which pinpoint influential signals to 20 ms specificity.
- BBC-sponsored project, which achieved state-of-the-art results on the FakeAVCeleb dataset.
- Conducted extensive data augmentation to mimic real-world detection evasion tactics and evaluated effects on model performance.
- Implemented experiments with HuggingFace and PyTorch and trained on Microsoft Azure ML virtual machines. See [thesis here](#) and [ICLR submission here](#).

Spectral DefocusCam | *Compressive Hyperspectral Imaging from Defocus Measurements*

Jun. 2021 – Aug. 2021

- Project associated with NSF Fellowship with Berkeley's Computational Imaging Lab.
- Optimized and democratized **remote sensing** by designing a tunable lens with a rapidly changing focus to reconstruct single-dimensional images to 31-dimensional **hyperspectral volumes**. See [poster here](#). Published in AAAI-22.

Bachelor's Thesis | *Machine Learning for Cardiac Arrhythmia Prevention*

Feb. 2019 – Jul. 2019

- Used respiratory rates to predict cardiac arrhythmia in hospital patients. Programmed with SQL, Python, and sci-kit learn to implement a Random Forest Classification model with the MIMIC-III database. The model achieved, at its best, an accuracy of 0.98 and F1-score of 0.97 in the prediction of cardiac arrhythmia.
- Received a score of 5.75 out of 6.
- Supervised by Dr. Walter Karlen and Dr. Patrick Schwab at ETH Zürich. See [code here](#) and [paper here](#).

AWARDS & HONORS

Autonomous Intelligent Machines & Systems (AIMS) Scholar

United Kingdom Research and Innovation Council

May 2024

Google DeepMind Scholarship

Oxford Department of Computer Science

May 2023

Graduate Research Fellowship (NSF GRFP)

National Science Foundation - declined

Mar. 2023

Grace Hopper Celebration Speaker

Association of Computer Machinery

Sep. 2022

Excellence in Undergraduate Research

University of Tennessee

May 2022

AAAI-22 Undergraduate Consortium Scholar

Association for the Advancement of Artificial Intelligence

Feb. 2022

NSF Research Experience for Undergraduates

University of California, Berkeley

Summer 2021

Grace Hopper Scholar

Association of Computer Machinery

Sep. 2021

PUBLICATIONS & TALKS

- [1] **Georgia Channing**, Juil Sock, Ronald Clark, Philip Torr, and Christian Schroeder de Witt. “Toward Robust Real-World Audio Deepfake Detection: Closing the Explainability Gap”. In: 2024. arXiv: 2410.07436.
- [2] **Georgia Channing**, Ria Patel, Ariel Rorabaugh, Paula Olaya, Silvina Caino-Lores, Catherine Schuman, Osamu Miyashita, Florence Tama, and Michela Taufer. “Composable Workflow for Accelerating Neural Architecture Search Using In Situ Analytics for Protein Characterization”. In: *Proceedings of the 52nd International Conference on Parallel Processing (ICPP)*. ACM, Aug. 2023, pp. 1–10.
- [3] **Georgia Channing**, Ria Patel, Ariel Rorabaugh, Paula Olaya, Silvina Caino-Lores, Catherine Schuman, Osamu Miyashita, Florence Tama, and Michela Taufer. *Generating Efficient Neural Networks for Protein Diffraction Data*. Project talk for JLESC15 workshop. Joint Laboratory for Extreme Scale Computing (JLESC), Mar. 2023.
- [4] **Georgia Channing**. “Spectral DefocusCam: Compressive Hyperspectral Imaging from Defocus Measurements”. In: *Proceedings of the 36th AAAI Conference on Artificial Intelligence*. June 2022, pp. 13128–13129.
- [5] **Georgia Channing** and Catherine Schuman. *Strategies for Recruitment and Retention of Women in CS*. Project talk at GHC22. Grace Hopper Conference (ACM), Sept. 2022.
- [6] Ria Patel, Ariel Rorabaugh, Paula Olaya, Silvina Caino-Lores, **Georgia Channing**, Catherine Schuman, Osamu Miyashita, Florence Tama, and Michela Taufer. “A Methodology to Generate Efficient Neural Networks for Classification of Scientific Datasets”. In: *Proceedings of the IEEE International Conference on e-Science 18* (Oct. 2022), pp. 1–2.
- [7] **Georgia Channing**. “Predictive Power of Common Risk Factors for Cardiac Arrhythmias in Critical Care”. Swiss Federal Institute of Technology in Zürich (ETH Zürich), July 2019.

SKILLS

Foreign Languages: German (C1), Mandarin (C1), Russian (B2), Spanish (B2)

Coding Languages: Python, C/C++, SQL (Postgres), PySpark

Memberships: Society of Women Engineers (SWE), Institute of Electrical and Electronics Engineers (IEEE), Association for Computing Machinery (ACM)